Patch-space Beltrami denoising of 3D point clouds

Aaron Wetzler* and Guy Rosman† and Ron Kimmel†

*Department of Electrical Engineering, Technion
†Department of Computer Science, Technion

Problem statement

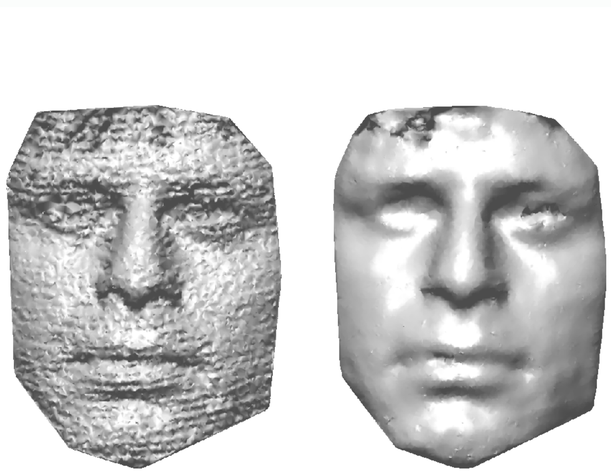
The Beltrami framework [1] treats images as two dimensional manifolds embedded in a hybrid spacial-spectral five dimensional $\{x, y, R, G, B\}$ space. Denoising color images can then be effectively performed through mean curvature area minimization of this embedded surface.

In [5][4] this approach has been efficiently extended to take advantage of neighboring image pixels. Can we now extend the patch-space based Beltrami method to denoise unstructured point clouds?

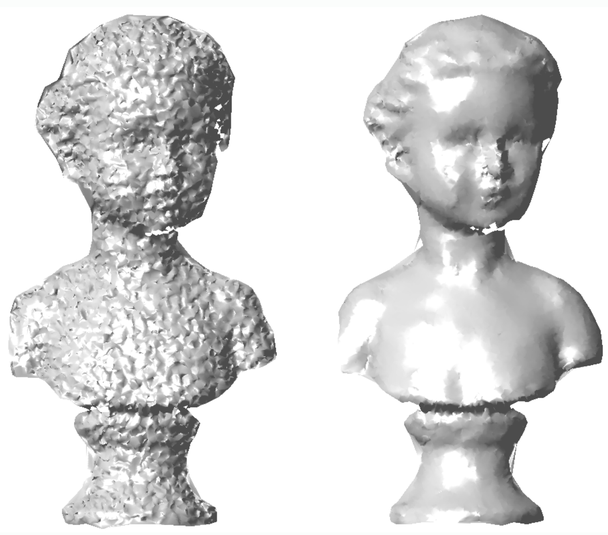
Contribution

- Extension of the Patch-Beltrami method onto unstructured point clouds.
- Exploration of how to define patches on point cloud data in the same spirit as [3].
- Investigation of the proposed method's denoising capacity when using robust normal estimation [2] as opposed to PCA.

Results

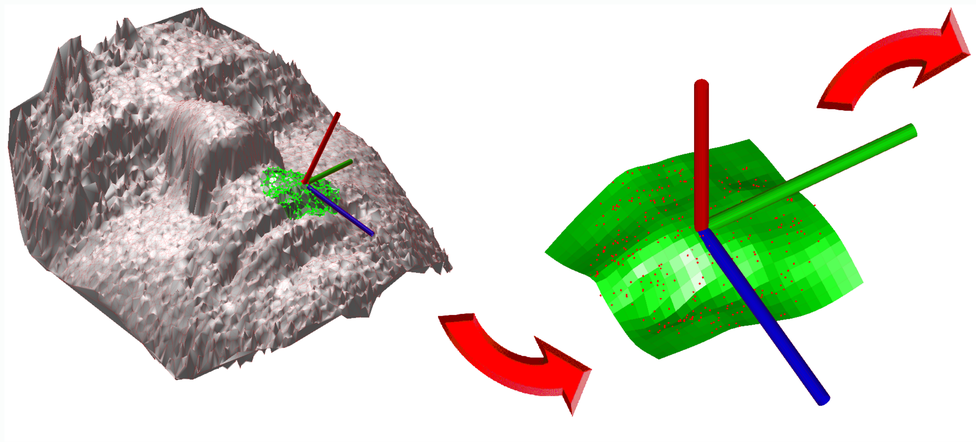


a) A point cloud obtained from a structured light scanner and smoothed using the proposed process. *Note that the clouds here have been triangulated for the sake of visual clarity however the denoising process makes no use of this structure.



b) The classic Bust model with additive artificial white noise denoised by the proposed process. The main visual structures have been preserved while smoothing the surface and removing artefacts introduced by the noise .

Processing pipeline

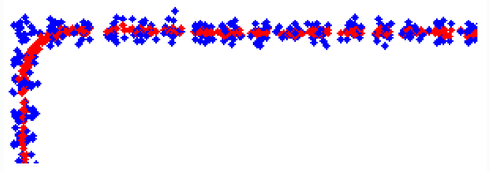


a) This image depicts the neighbourhood found around a vertex and the resulting coordinate frame determined from these points.

b) Once a support set and coordinate frame have been determined they are transformed into a canonical frame and a function I is interpolated over a grid.

c) During each denoising iteration the update is projected onto the vertex normals and all the vertices are updated simultaneusly after passing over the whole cloud.

Local frame estimation: PCA vs robust normals



The upper point set has been processed with PCA frame extraction and the lower point set with robust frame estimation [2]

The efficacy of this method is largely dependent on how well we can estimate the normals at every point. PCA tends to smooth out corners despite the Beltrami filter's edge preserving flow qualities. The method of Fleishman et al. [2] for robust normal estimation proved to be more reliable when denoising structures with sharp edges as can be seen in the image on the left.

References

[1] N. Sochen, R. Kimmel and R. Malladi A general framework for low level vision, In *IEEE Trans. on Image Processing '98*

[2] S. Fleishman and D. Cohen-Or and C. T. Silva Robust moving least-squares fitting with sharp features, In *SIGGRAPH '05*

[3] G. Rosman and A. Dubrovina and R. Kimmel Patch-collaborative spectral surface denoising, Technical report CIS-2012-03, Technion, '12

[4] A. Wetzler and R. Kimmel Efficient Beltrami flow in patch-space, In *SSVM '11*

[5] A. Roussos and P. Maragos, Tensor-based image diffusions derived from generalizations of the total variation and Beltrami functionals In *ICIP '10*

Beltrami framework

Functional

Minimizing the functional

$$S[X, G] = \iint \sqrt{g} dx dy$$

leads to an Euler-Lagrange equation which can be compactly written as

$$-div\left(\sqrt{g} G^{-1} \nabla I\right) = 0.$$

Freedom of parameterization permits appending $g^{-1/2}$ which yields the second order differential operator of Beltrami

$$\Delta_g I = 0.$$

Metric

Equating distances on the manifolds produces the induced metric tensor

$$G = \begin{pmatrix} 1 + I_x^2 & I_x I_y \\ I_x I_y & 1 + I_y^2 \end{pmatrix}.$$

Point cloud patch-space metric

Patch definition

We extract a neighborhood of N neighbors that lie within a radius

$$R = max\left(5\sigma, \delta\right)$$

where δ is the average space between vertices. The neighbors are rotated to a canonical frame and a function I is generated by interpolating the data over a fixed grid. Over the function I we define a patch of size w×w as

$$P(x, y) = \left(x, y, \left\{I^k\left(x + iw, y + jw\right)\right\}\right)$$

where $i, j = -w, .., w,\ k = 1, .., n.$

Patch-space metric

Using this extended feature space we derive the induced metric tensor as

$$G = \begin{pmatrix} 1 + \beta^2 \sum_{i,j,k} I^{k\,2}_{i,j\,x} & \beta^2 \sum_{i,j,k} I^k_{i,j\,x} I^k_{i,j\,y} \\ \beta^2 \sum_{i,j,k} I^k_{i,j\,x} I^k_{i,j\,y} & 1 + \beta^2 \sum_{i,j,k} I^{k\,2}_{i,j\,y} \end{pmatrix}$$

Vertex update

Taking the central pixel we determine

$$I_t = \Delta_g I = \frac{1}{\sqrt{g}} div\left(\sqrt{g} G^{-1} \nabla I\right).$$

A vertex update along the patch's normal using only the iterated change of the center patch pixel is performed.